Analysis Report

Instructor : Dr. Islam El-gedawy

Student name : Tasneem Ibrahim Hassan Mohamed Taha (PM)
Abdulrahman Hamby Khalil Shehata Harb
Rokia Islam Elsayed Hussein Elsayed Ali
Mariam Mohamed Sobhy Mohamed Othman
Raghad Mohamed Ahmed Mohamed Abu Rafai

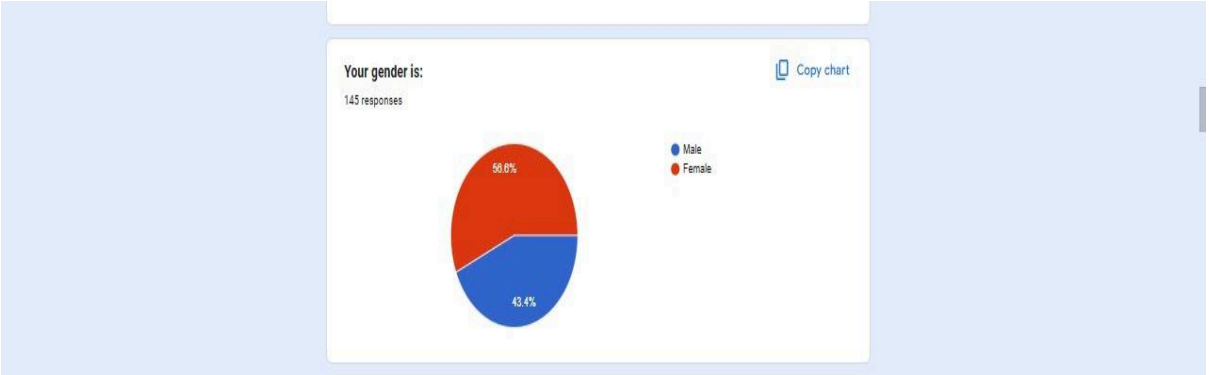
Student's ID : 22101382
22100692
22101084
22100778
22101912

Faculty : Computer science and engineering (CSE)

Level : Three Semester : Fall 2025

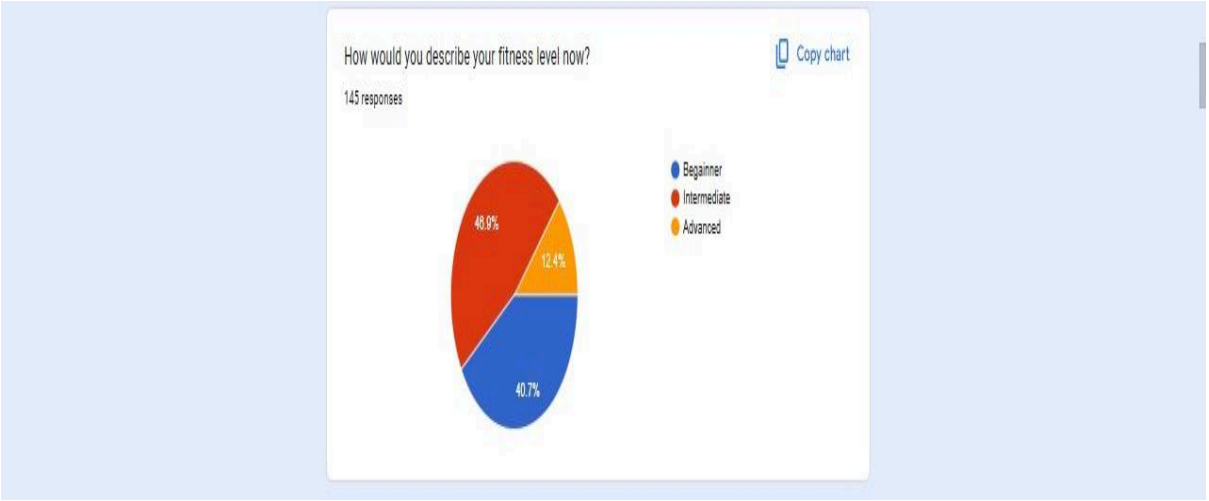
Course name : Software Engineering Course code : CSE251

Figure 1:



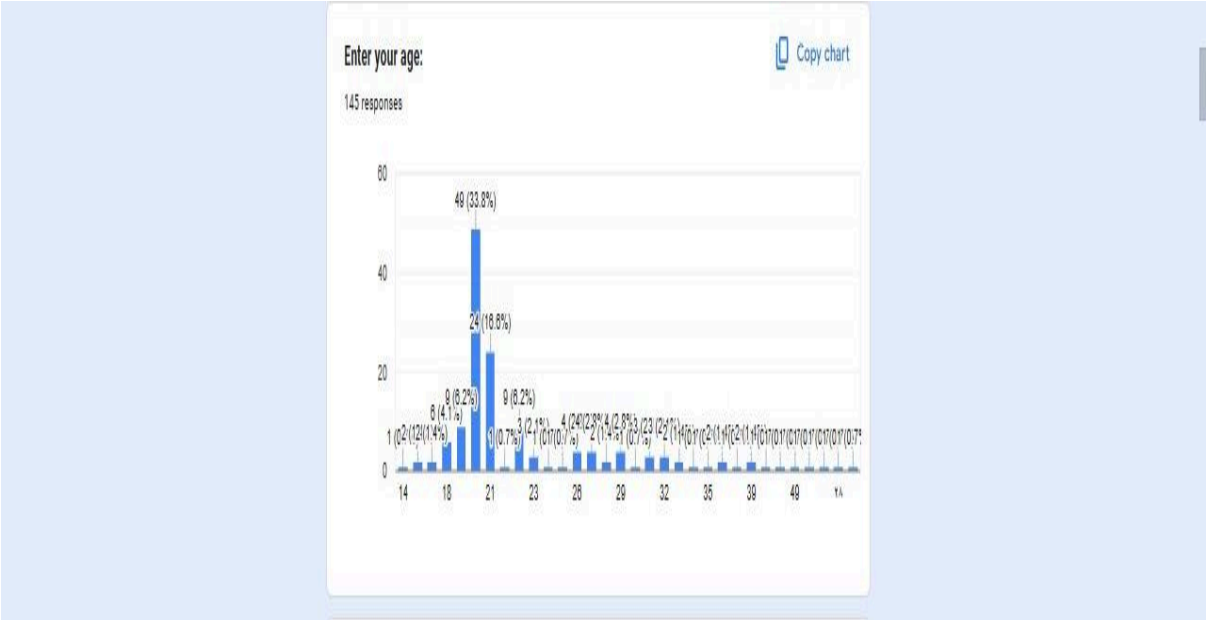
The majority is female respondents which concludes that females are more focused on their fitness level than men as they tend to care more about their looks and appearances

Figure 2:



As shown in the diagram
Intermediate fitness level is the majority so the app needs to have a big focus on that by offering various workouts for them and for beginner level it has also a larger percentage and the app needs to on leveling up that percentage to intermediate level but as for advanced it’s the least percentage

Figure 3:

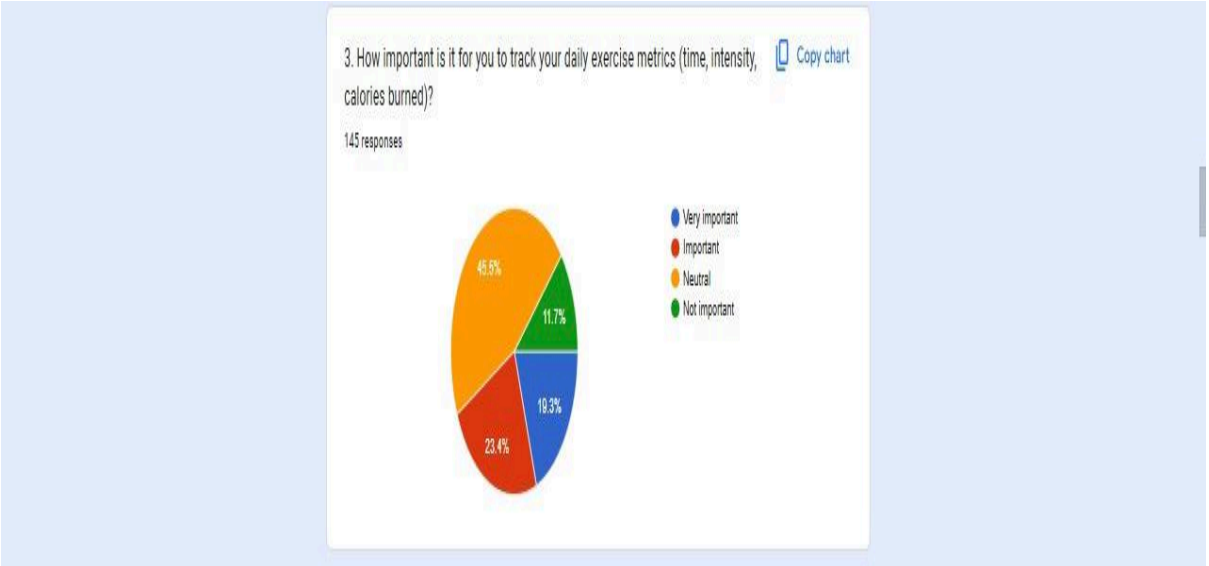


Here is the numerical analysis based on the estimated data:

Mean Age: 22.6 years
Median Age: 19 years
Mode Age: 18 years (most frequent age with 49 responses)
Percentage Distribution (Main Ages 18-21):
Age 18: 37.12%
Age 19: 18.18%
Age 20: 6.82%
Age 21: 6.82%

Age 18: 37.12% — the largest single age group, indicating that the population is heavily skewed towards the youngest individuals.
Age 19: 18.18% — a significant portion, although notably less than age 18.
Ages 20 and 21: Each contributes 6.82%, reflecting much smaller representation in slightly older individuals.

Figure 4:

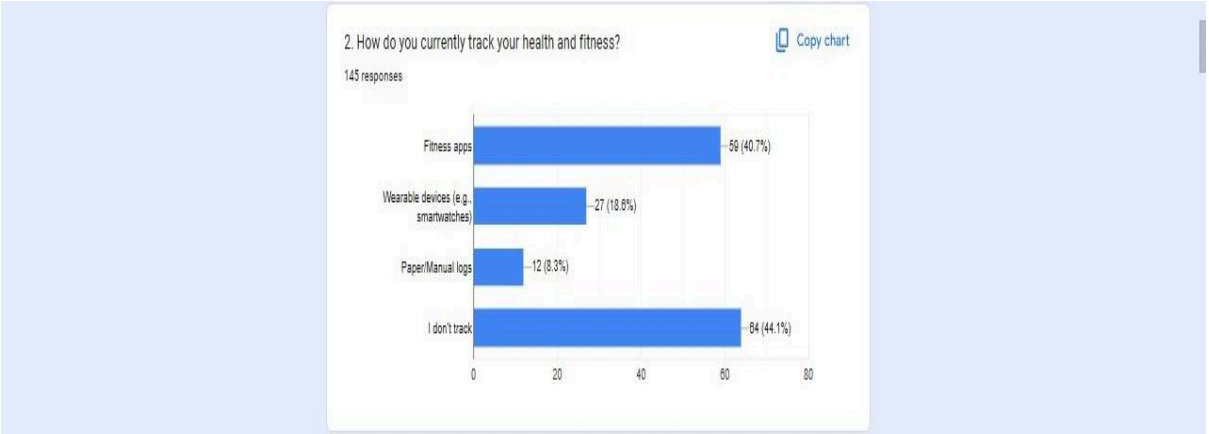


Here is the numerical analysis based on the survey results:
Very Important: 66 responses (45.5%)
Important: 34 responses (23.4%)
Neutral: 28 responses (19.3%)
Not Important: 17 responses (11.7%)

High Overall Importance:

- **45.5% (66 responses)** consider it **"Very Important"** to track exercise metrics, and an additional **23.4% (34 responses)** find it **"Important"**.
    - Combined, **68.9%** of respondents view tracking exercise metrics positively.
    - This indicates that the majority of people see clear value in monitoring their fitness progress, suggesting it is a priority for most.

Figure 5:



From the chart:

Numerical Analysis:
- *Fitness Apps:*
  - Responses: 59
  - Percentage: 40.7%

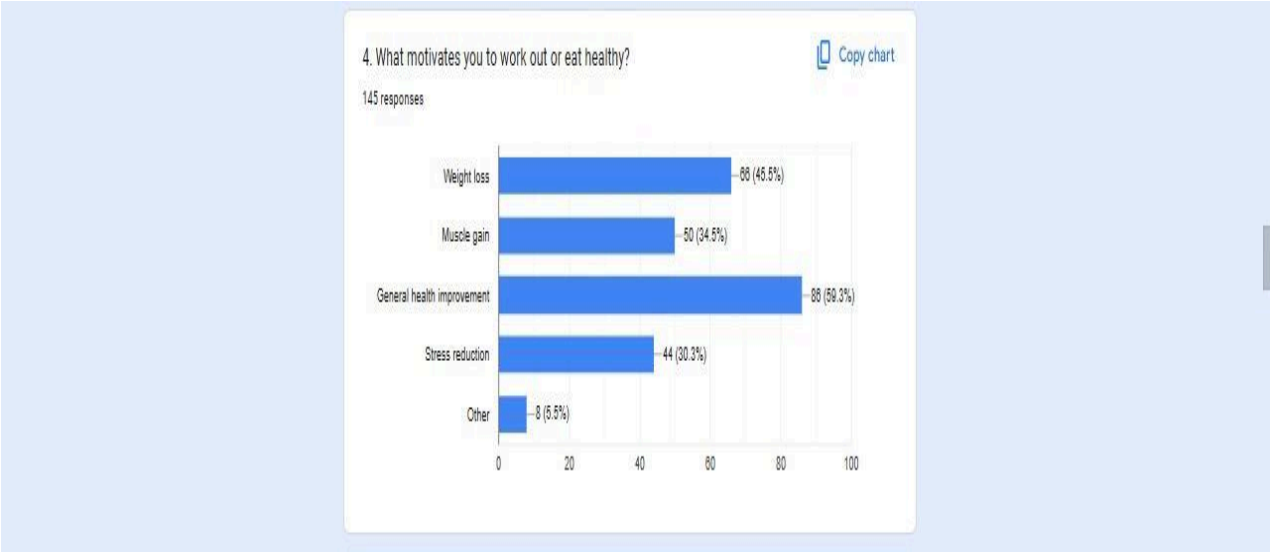
- *Wearable Devices (e.g., Smartwatches):*
  - Responses: 27
  - Percentage: 18.6%

- *Paper/Manual Logs:*
  - Responses: 12
  - Percentage: 8.3%

- *I Don’t Track:*
  - Responses: 84
  - Percentage: 44.1%

Key Insights:
- *Majority don't track their health and fitness (44.1%).*
- *Fitness apps are the most popular tracking method (40.7%).*
- *Wearable devices are used by 18.6% of respondents.*
- *Paper/manual logging is the least preferred (8.3%).*

Figure 6:



From the chart:

Numerical Analysis:
- *Weight Loss:*
  - Responses: 66
  - Percentage: 45.5%

- *Muscle Gain:*
  - Responses: 50
  - Percentage: 34.5%

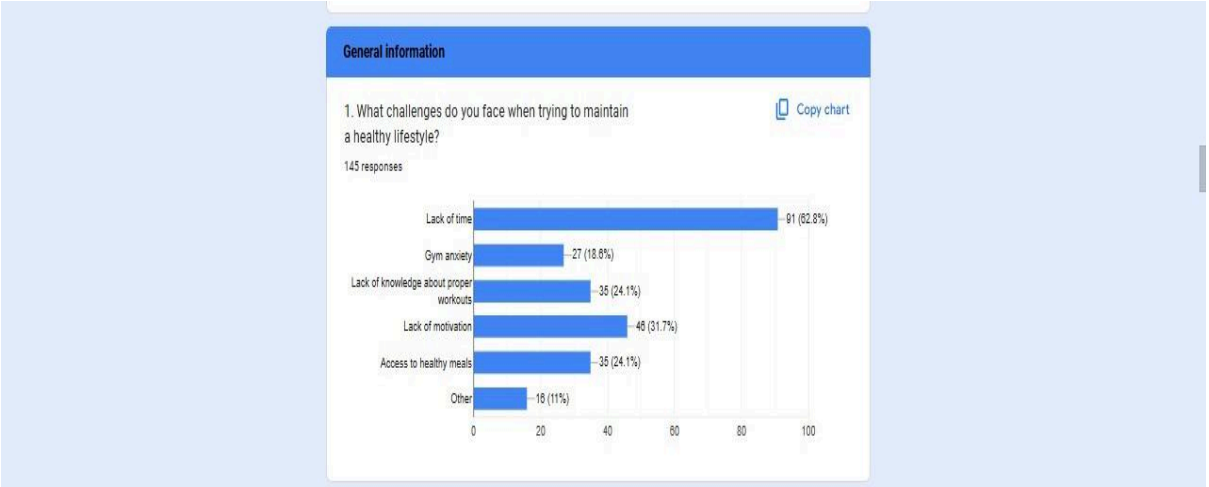
- *General Health Improvement:*
  - Responses: 88
  - Percentage: 60.7%

- *Stress Reduction:*
  - Responses: 44
  - Percentage: 30.3%

- *Other:*
  - Responses: 8
  - Percentage: 5.5%

Key Insights:
- *General health improvement is the top motivator (60.7%).*
- *Weight loss follows as a significant motivator (45.5%).*
- *Muscle gain motivates a smaller, yet substantial portion (34.5%).*
- *Stress reduction influences 30.3% of respondents.*
- *Other motivations are relatively minimal (5.5%).*

Figure 7:



Observations:
Top Challenge:

Lack of time is the most common challenge, affecting 62.8% of respondents (91 votes).
Second Most Common Challenge:

Lack of motivation received 31.7% (46 votes).
Moderately Common Challenges:

Lack of knowledge about proper workouts and Access to healthy meals both received 24.1% (35 votes each).
Least Common Challenge:

Gym anxiety affects 18.6% (27 votes).
Other:

11% of respondents (16 votes) reported facing challenges not listed among the primary options.
Insights:
Time constraints are by far the biggest barrier to maintaining a healthy lifestyle.
Motivation also poses a significant challenge, whereas knowledge gaps, meal access, and gym anxiety are less widespread but still notable concerns.

Challenge	Percentage	No. of responses
Lack of time	62.8%	91
Gym anxiety	18.6%	27
Lack of knowledge about Proper workouts	24.1%	35
Lack of motivation	31.7%	46
Access to healthy meals	24.1%	35
Other	11%	16

The survey, with 145 responses, identifies the main challenges individuals face when trying to maintain a healthy lifestyle.

Top Challenge:

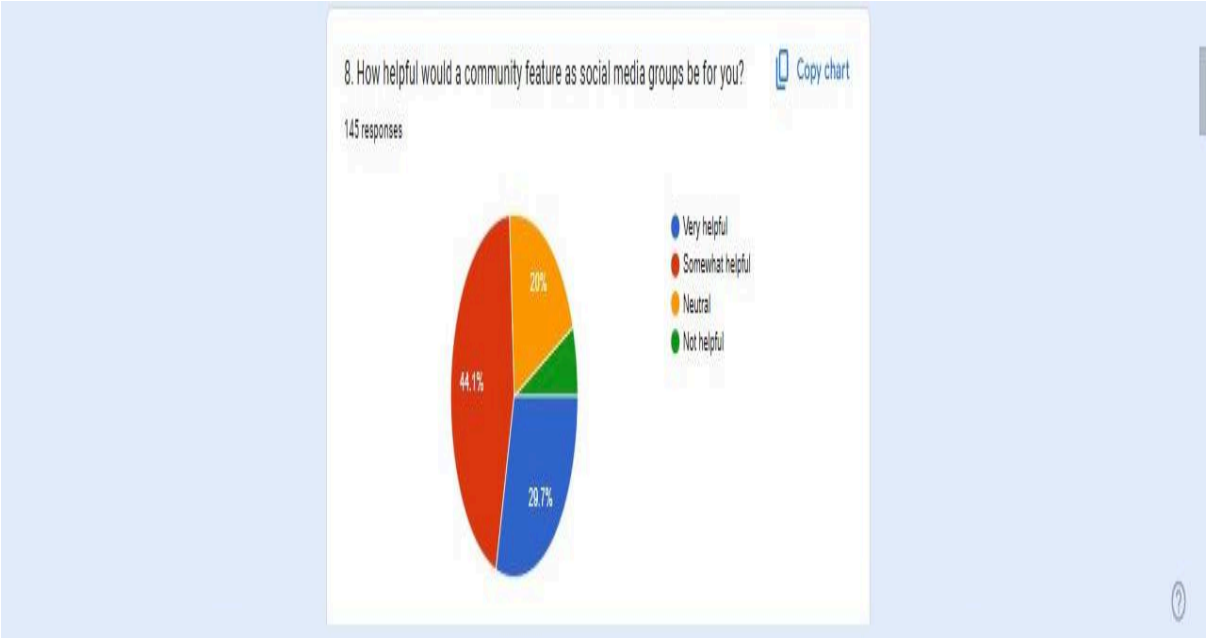
Lack of time is the most significant challenge, with 91 votes (62.8%), showing that busy schedules are a major barrier.
Moderate Challenges:

Lack of motivation was chosen by 46 respondents (31.7%), indicating struggles with staying consistent and driven.
Both lack of knowledge about proper workouts and access to healthy meals received 35 votes (24.1%), highlighting gaps in understanding exercise routines and obtaining nutritious food.
Lower Challenge:

Gym anxiety was reported by 27 respondents (18.6%), showing a smaller but notable segment struggles with comfort in fitness spaces.
Other:

16 votes (11%) reflect additional personal or unique challenges.
Conclusion:
The data reveals that time constraints are the most critical barrier, followed by motivation and knowledge gaps. Addressing these issues, such as through time-efficient routines, motivation tools, and educational resources, could help users better maintain a healthy lifestyle.

Figure 8:



Here's a *detailed numerical analysis* based on the provided chart:

Total Responses: *145*

Breakdown by Category:
1. *Very Helpful*
   - *Number of Responses:* 42
   - *Percentage of Total:* $\frac{42}{145} \times 100 = 29.7\%$
   - Insight: Nearly one-third of respondents find the feature very beneficial.

2. *Somewhat Helpful*
   - *Number of Responses:* 64
   - *Percentage of Total:* $\frac{64}{145} \times 100 = 44.1\%$
   - Insight: This is the largest group, indicating moderate interest or partial support for the feature.

3. *Neutral*
   - *Number of Responses:* 29
   - *Percentage of Total:* $\frac{29}{145} \times 100 = 20.0\%$
   - Insight: A fifth of respondents are indifferent, meaning this group may be open to persuasion depending on how the feature is presented.

4. *Not Helpful*
   - *Number of Responses:* 10
   - *Percentage of Total:* $\frac{10}{145} \times 100 = 6.9\%$
   - Insight: A small minority does not see value in this feature.

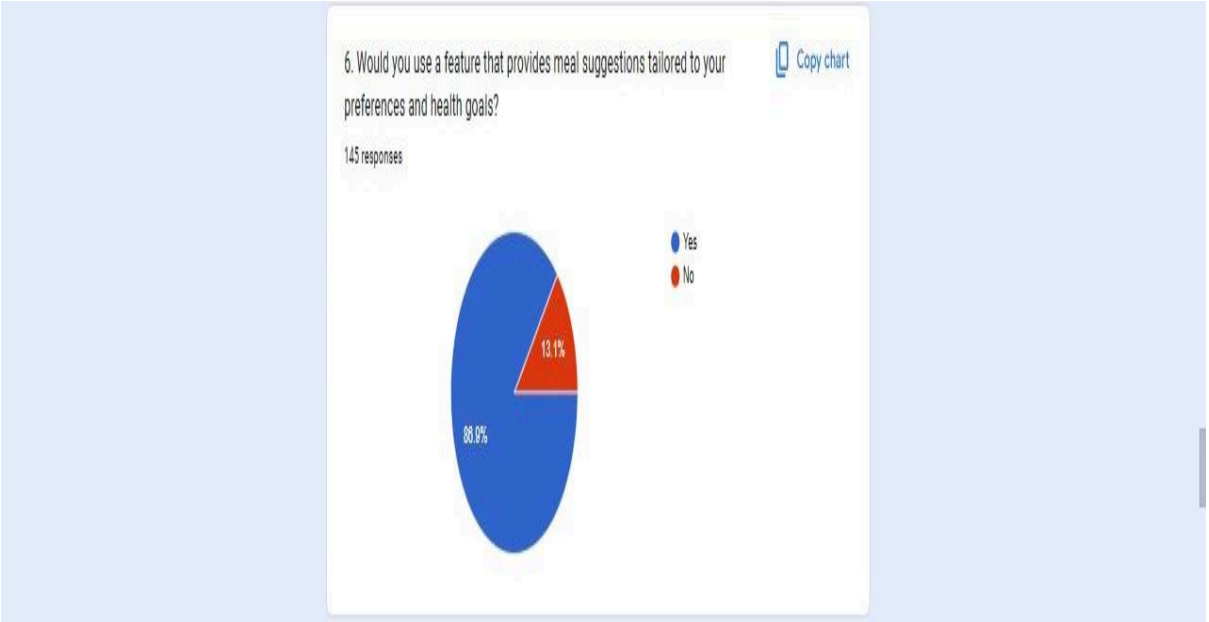
Additional Observations:
1. *Positive Perception (Very Helpful + Somewhat Helpful):*
   - Combined Responses: $42 + 64 = 106$
   - Combined Percentage: $\frac{106}{145} \times 100 = 73.8\%$
   - Insight: Over 70% of respondents see value in the feature to some degree, showing strong potential acceptance.

2. *Negative Perception (Not Helpful):*
   - Responses: 10
   - Percentage: 6.9%
   - Insight: The opposition is minimal, indicating low risk in implementing the feature.

3. *Indifference (Neutral):*
   - Responses: 29
   - Percentage: 20.0%
   - Insight: This group could be influenced positively with clearer benefits and marketing.

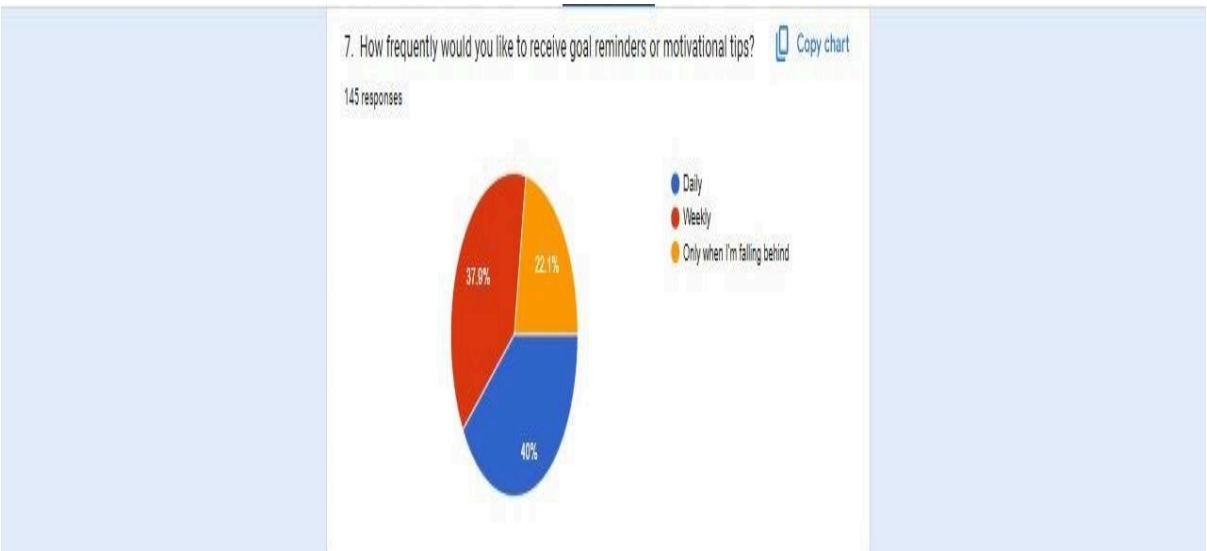
Comparison:
- The *Somewhat Helpful* category has the highest response count at 44.1%, showing partial but dominant support.
- The *Very Helpful* group at 29.7% provides substantial additional support.
- Only 6.9% of respondents consider the feature *Not Helpful*, a negligible minority.
- The *Neutral* group at 20% suggests room for improvement in communicating the feature's value.

Figure 9:



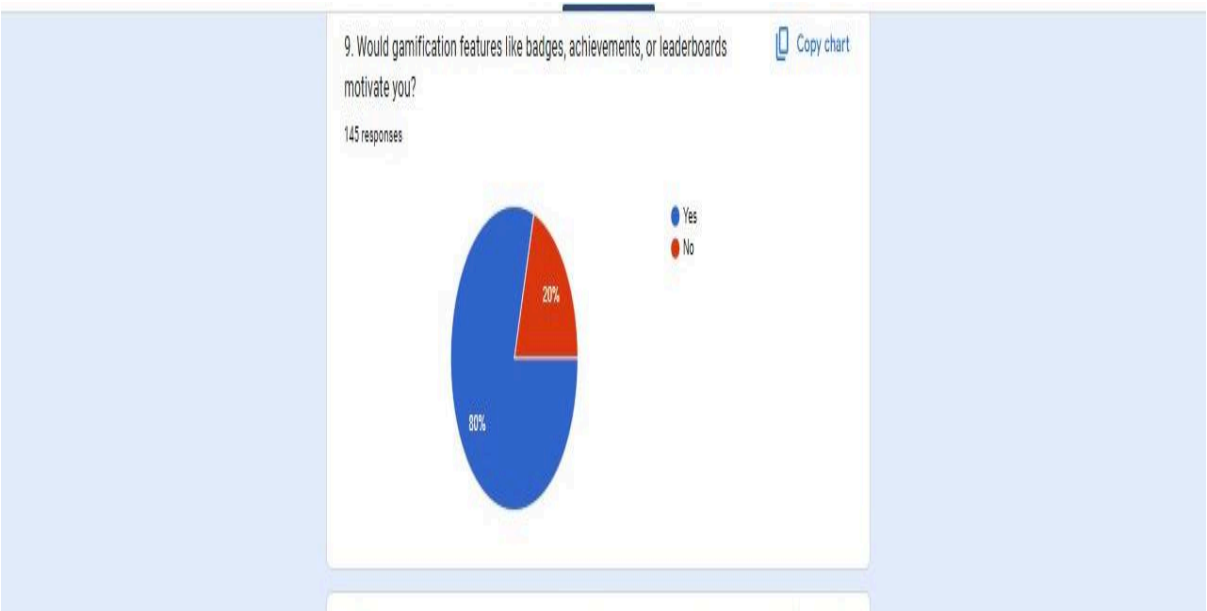
The pie chart indicates a strong positive response to the idea of a meal suggestion feature tailored to user preferences and health goals within a fitness app. A significant majority (86.9%) of respondents expressed willingness to use such a feature. so the app needs to have a great focus on such a feature and have regular updates for it

Figure 10:



Daily Notifications: 49% of users prefer daily notifications, indicating a strong need for regular reminders or updates. This segment might benefit from features like daily workout challenges or habit-tracking reminders.
Weekly Notifications: 26.2% of users prefer weekly notifications, suggesting they value a less frequent, more summarized approach. This group might benefit from weekly progress reports or motivational messages.
Only When Necessary: 20.7% of users prefer notifications only when necessary. This indicates a desire for control and a preference for receiving information only when relevant. This segment might benefit from features like personalized notifications based on their activity or goals.
Never: a small percentage of users prefer no notifications at all. This suggests a strong preference for self-directed engagement and a desire to avoid distractions. This group might benefit from features that allow them to manually trigger notifications or reminders.

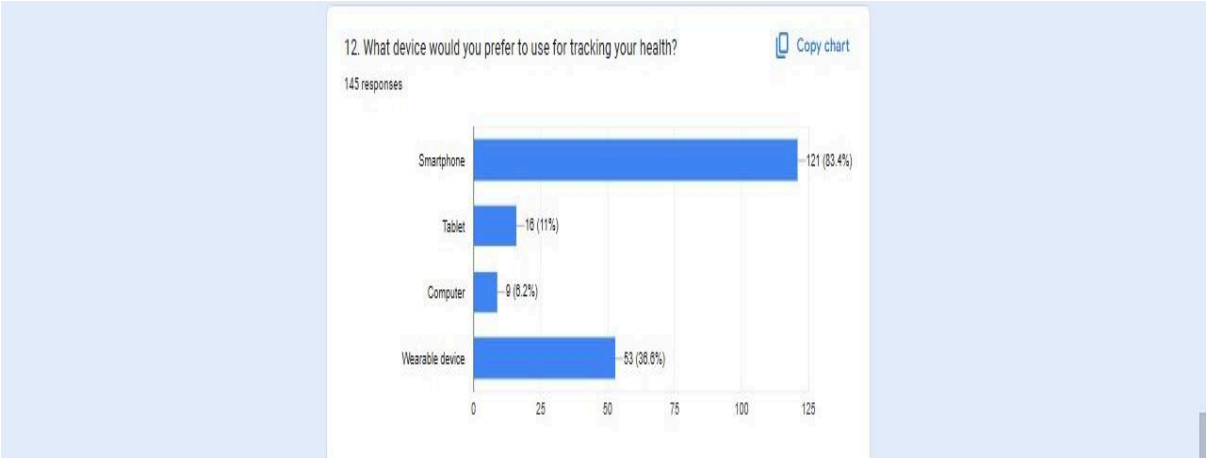
Figure 11:



Here is the numerical analysis for the survey question about gamification features:

Yes: 116 responses (80%)
No: 29 responses (20%)
So the app needs to give this feature such a great focus as it so important and motivating for it is population and 80% want it

Figure 12:



The image contains a bar chart showing responses to the question: “What device would you prefer to use for tracking your health?” Here’s a numerical analysis based on the chart:

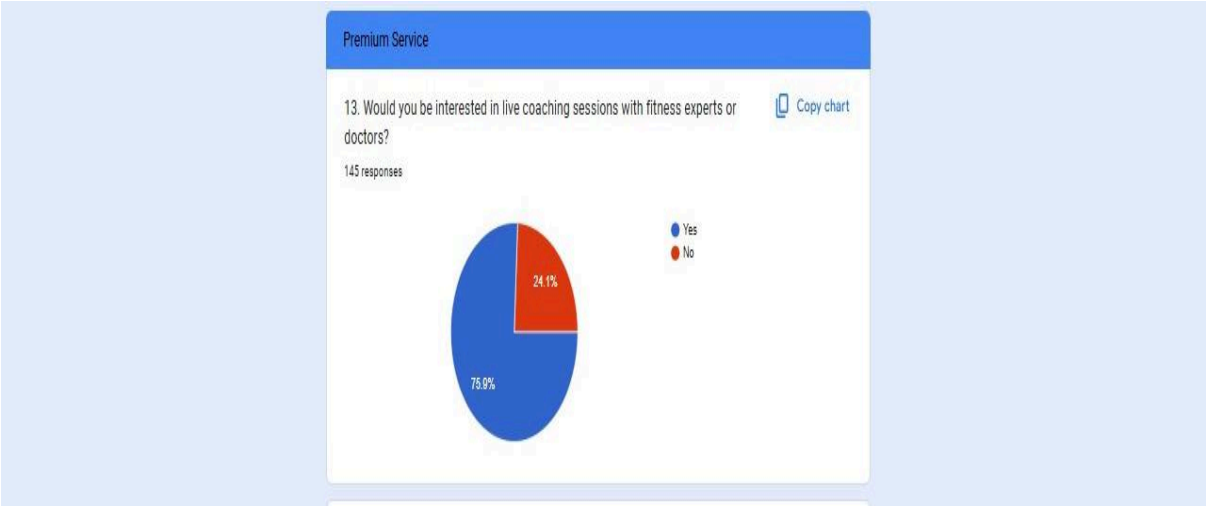
1. Smartphone
• Responses: 121
• Percentage: 83.4%
2. Tablet
• Responses: 16
• Percentage: 11%
3. Computer
• Responses: 9
• Percentage: 6.2%
4. Wearable Device
• Responses: 53
• Percentage: 36.6%

Total Responses: 145

Key Observations:
• The majority (83.4%) of respondents prefer smartphones for health tracking.
• Wearable devices are the second most preferred, with 36.6% of respondents.
• Tablets and computers are less popular, with 11% and 6.2% preferences, respectively.

The data reveals that smartphones are the most preferred device for health tracking (83.4%), followed by wearable devices (36.6%), highlighting the importance of portability and convenience. Tablets (11%) and computers (6.2%) are less favored, likely due to their limited practicality. The trend emphasizes a strong preference for mobile and wearable technology in health monitoring.

Figure 13:



Observations

1. Total Responses: 145 (as stated on the chart).
2. Category Distribution:

- Yes: 75.9%
- No: 24.1%

Detailed Numerical Analysis

1. Yes

- Percentage: 75.9%
- Number of responses:

145 \times 75.9\% = 145 \times 0.759 = 110.055 \approx 110

2. No

- Percentage: 24.1%
- Number of responses:

145 \times 24.1\% = 145 \times 0.241 = 34.945 \approx 35

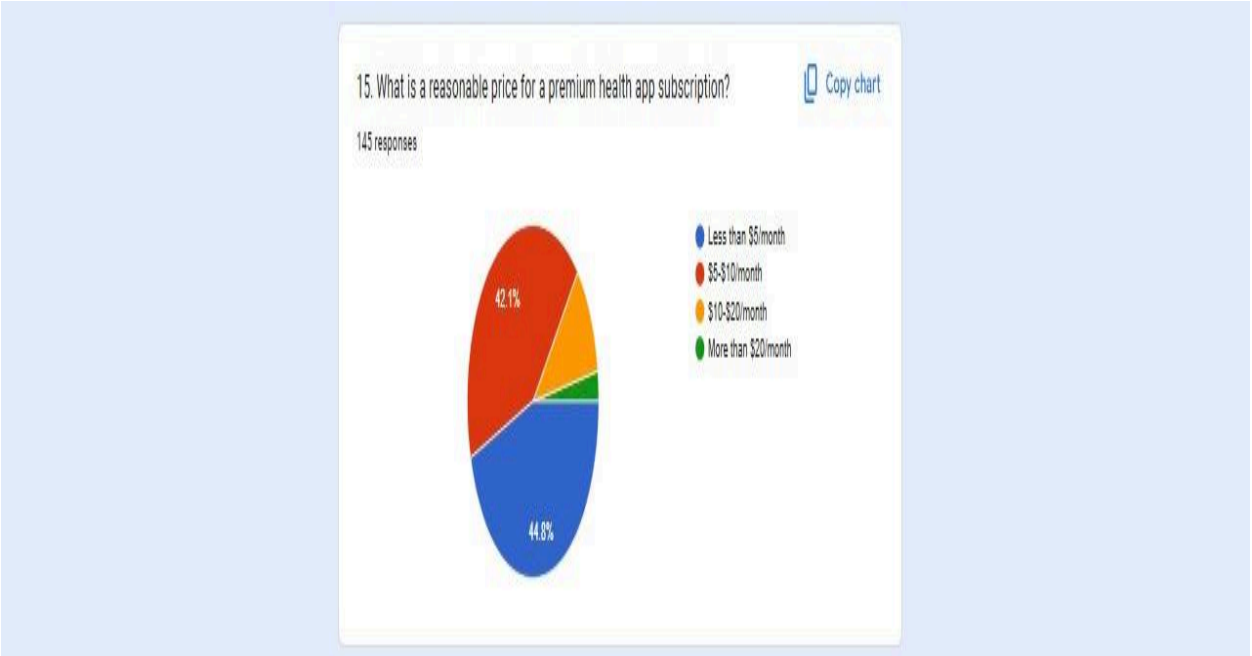
Insights:

- A significant majority

Response	percentage	No. of response
Yes	75.9%	110
NO	24.1%	35

the majority of respondents (75.9%, or 110 out of 145) expressed interest in live coaching sessions with fitness experts or doctors, indicating strong demand for personalized, interactive health and fitness support. A smaller portion (24.1%, or 35 respondents) were not interested, suggesting that while most users value this feature, some may prefer other forms of engagement or do not see it as a necessary service.

Figure 14:



Observations

1. Total Responses: 145 (as stated on the chart).
2. Category Distribution:

- Less than $5/month: 44.8%
- $5–$10/month: 42.1%
- $10–$20/month: Estimated around 10% (orange segment).
- More than $20/month: Estimated around 3% (green sliver).

Detailed Numerical Analysis

1. Less than $5/month
    - Percentage: 44.8%
    - Number of responses:

145 \times 44.8\% = 145 \times 0.448 = 65

2. $5–$10/month
    - Percentage: 42.1%
    - Number of responses:

145 \times 42.1\% = 145 \times 0.421 = 61

3. $10–$20/month
    - Estimated Percentage: 10%
    - Number of responses:

145 \times 10\% = 145 \times 0.10 = 15

4. More than $20/month
    - Estimated Percentage: 3%
    - Number of responses:

145 \times 3\% = 145 \times 0.03 = 4

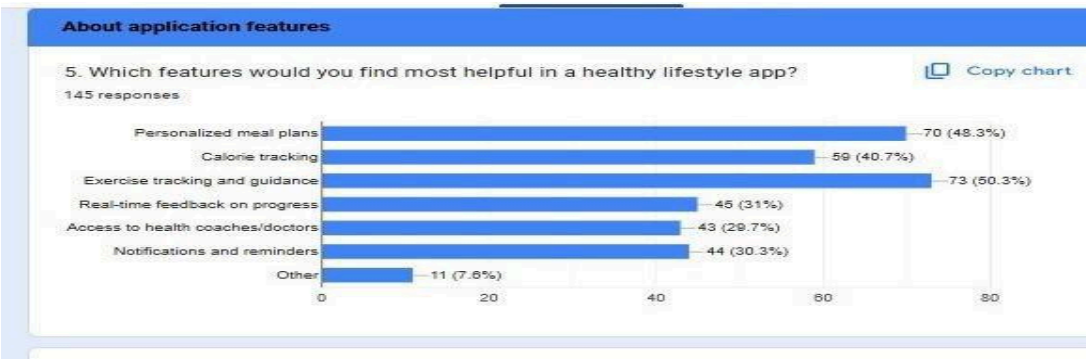
Insights:

- The majority (approximately 87% of respondents) are willing to pay $10 or less per month for a premium health app.
- A very small portion of the audience (3%) is open to paying more than $20/month

the majority of respondents (86.9%, or 126 out of 145) consider a price of $10 or less per month reasonable for a premium health app subscription, with 44.8% favoring less than $5 and 42.1% favoring $5–$10. A smaller group (10%, or 15 respondents) is willing to pay $10–$20 per month, while only 3% (4 respondents) are open to paying more than $20. This suggests that affordability is a key factor for most users when evaluating premium health app subscriptions.

Price Range	percentage	No. of responses
Less than 5$/month	44.8%	65
5-10/month	42.1%	61
10-20/month	10%	15
More than 20$/month	3%	4

Figure 15:



Observations:

Top Feature:

Exercise tracking and guidance is the most popular feature with 50.3% of respondents (73 votes).
Second Most Popular Feature:

Personalized meal plans received 48.3% (70 votes).
Third Most Popular Feature:

Calorie tracking comes third with 40.7% (59 votes).
Least Selected Feature:

Other is the least selected, with only 7.8% (11 votes).
Insights:
Over half of the respondents value exercise tracking and personalized meal plans.
Health coaching, notifications, and real-time feedback have similar lower interest levels, each receiving around 30% of votes.

Feature	Number of responses	Percentage
Personalized meal plans	70	48.3
Calorie tracking	59	40.7
Exercise tracking and guidance	73	50.3

Real time feedback on progress	45	31
Access to health coaches/doctors	43	29.7
Notification and reminders	44	30.3
other	11	7.8

Analysis of Healthy Lifestyle App Features Survey:

The survey, consisting of 145 responses, highlights the most helpful features users seek in a healthy lifestyle app.

Top Features:

Exercise tracking and guidance is the most favored feature, with 73 votes (50.3%). Personalized meal plans follow closely at 70 votes (48.3%), showing significant interest in tailored nutrition solutions.
Moderately Popular Features:

Calorie tracking ranks third with 59 votes (40.7%), indicating a notable preference for tools that monitor food intake.
Real-time feedback on progress received 45 votes (31%), reflecting user interest in performance tracking.
Lower Preference:

Access to health coaches/doctors and notifications/reminders had 43 votes (29.7%) and 44 votes (30.3%), respectively.
Other features received minimal interest, with 11 votes (7.8%).
Conclusion:
The findings indicate a strong demand for exercise guidance, meal personalization, and calorie tracking. Features like notifications, access to health professionals, and real-time progress feedback hold moderate importance but are secondary priorities.